"Empowerment through quality technical education"

Ajeenkya DY Patil School of Engineering

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Department of Computer Engineering

Assignments **Form No. IQAC/18a1**

Class: SE Div: C	Course Name: Data Structure	Course Coordinator: Prof.Ishwar Bharambe
Assignment No.2	Assignment based on Units: 2	Date of issue: 31/08/2026

CO 2: Apply various sorting and searching algorithms for given problem

Sr. No.	Questions	Level of Mapping asnd Number			Blooms Level
		CO	PO	PSO	
S.N.	**Question**	**CO**	**PO**	**PSO**	**Bloom's Level**
Q.1.	Describe the operations performed on an array. How is a multidimensional array represented in memory?	CO2	PO1, PO2	PSO1	2
Q.2.	Write and analyze the algorithm for Binary Search and Fibonacci Search. Compare their efficiencies.	CO2	PO2, PO4, PO5	PSO1	4
Q.3.	Explain the concept of Sparse Matrix and its storage representation.	CO2	PO1, PO2	PSO1	2
Q.4.	Demonstrate the working of Quick Sort and Merge Sort with examples and discuss their time complexities.	CO2	PO1, PO2, PO4, PO5	PSO2	4
Q.5.	Analyze the importance of stability and efficiency in sorting algorithms. Which sorting technique is most suitable for large datasets and why?	CO2	PO2, PO5	PSO2	5

Sign of Course Coordinator:Prof.Ishwar Bharambe	Sign of HoD

